

Seating arrangements

COMPLETE LEARNING GUIDE

What You'll Learn

Seating arrangement questions extend puzzle-solving to a specific, heavily-tested format: linear rows, circular tables, and double rows, each with their own conventions for how 'left', 'right', 'facing', and 'opposite' should be interpreted. The single biggest source of wrong answers is applying linear-row intuition to circular or facing-inward/outward arrangements, where left and right flip depending on which way people face — a mechanical rule, not guesswork, resolves this correctly every time. This sub-module builds a consistent notation and diagramming habit, covers linear arrangements (including counting from left vs right), circular arrangements (facing center vs facing outward), double-row arrangements where two rows face each other, the critical distinction between 'left of' and 'immediately left of,' and how to work through negative ('is not') constraints — all through fully solved examples.

1. Linear Arrangement Basics

In a row of n people, 'kth from the left' and 'kth from the right' refer to different positions unless explicitly stated — position from the right = $n - \text{rank from right} + 1$. Always convert to one consistent numbering system before solving.

- 6 people sit in a row, all facing north. B sits 3rd from the left (position 3). E sits 2nd from the right — using position = $n - \text{rank} + 1 = 6 - 2 + 1 = 5$, E is at position 5. A sits immediately left of B, so A is at position 2.

2. Circular Arrangements: Facing Center vs Facing Outward

For people seated facing the CENTER of a circular table, moving clockwise corresponds to a person's LEFT and moving anticlockwise corresponds to their RIGHT — this feels counterintuitive but is the standard convention, and it fully reverses if the people instead face AWAY from the center.

- 6 people sit around a table facing the center, seats numbered 1–6 clockwise. Q is at seat 1. R is opposite Q (3 seats apart in a 6-seat circle) ! R is at seat 4. P is immediately to Q's right — since facing center, 'right' = anticlockwise — so P is at seat 6.

3. Diagramming Technique

Draw the row or circle as blank slots first, place every absolute-position clue, then work relative clues outward from those anchors — updating the diagram after each deduction (rather than trying to hold intermediate states mentally) prevents the single most common source of seating-arrangement errors: losing track of an earlier placement.

4. Double Row Arrangements

When two rows face each other (e.g., Row 1 facing south, Row 2 facing north), positions are typically compared using each row's own physical left-to-right numbering, and 'opposite' pairs the same position-numbers across rows — the rows facing each other means viewer-perspective left/right differs between the two rows, so always anchor to physical, not viewer, position numbers.

- Row 1 (facing south): P, Q, R, S at positions 1–4 (left to right). Row 2 (facing north): W, X, Y, Z, each seated opposite someone in Row 1. W is opposite Q (position 2) ! W is in Row 2's position 2. X is opposite S (position 4) ! X is in Row 2's position 4. Y and Z occupy positions 1 and 3, order undetermined without a further clue.

5. 'Left Of' vs 'Immediately Left Of'

'A is to the left of B' only means A is somewhere to B's left (any distance) — it does not fix adjacency. 'A is immediately to the left of B' fixes A directly next to B. Misreading one for the other is the single most common seating-arrangement error and can allow (or wrongly eliminate) many valid arrangements.

- In a row of 5, 'C is to the left of D' alone allows several arrangements (C could be in position 1, 2, 3, or 4 relative to D anywhere further right). 'C is immediately to the left of D' fixes them as an adjacent CD block, sharply reducing the possibilities.

6. Working Through Negative ('Is Not') Constraints

Negative clues ('X is not at an end', 'Y is not adjacent to Z') are best applied AFTER positive clues have narrowed the open slots, since a negative clue only eliminates options rather than fixing a position — applying them too early with too many open slots wastes time checking cases that get resolved anyway by later positive clues.

- 5 seats. H is at an end (seat 1 or 5). F is not at an end and sits exactly in the middle (seat 3). G is not adjacent to F, so G " seat 2 and G " seat 4 ! G must be seat 5 (if H=1) or seat 1 (if H=5). This narrows the remaining two people to the two leftover seats.

7. Common Traps in Direction-Based Seating

The exact same relative clue ('P is to the right of Q') produces a different physical arrangement depending on whether people face inward or outward, and whether the row faces the reader or faces away — always establish the facing direction from the question stem before applying any left/right clue, and re-derive the convention rather than assuming it matches a previous question's setup.

Common Pitfalls

- Applying viewer's left/right instead of the seated person's own left/right when a direction is not explicitly the reader's perspective.
- Forgetting that for people facing the CENTER of a circle, clockwise = their left and anticlockwise = their right (reversed for facing outward).
- Treating 'X is to the left of Y' as adjacency when only 'immediately to the left' guarantees that.
- In double-row arrangements, comparing viewer-perspective positions across the two rows instead of each row's own physical left-to-right numbering.
- Applying negative ('is not') constraints before positive/absolute constraints, wasting time on cases that get eliminated anyway.
- Not re-deriving the facing convention for each new question, carrying over an assumption from a previous, differently-configured puzzle.

Key Takeaways

- Position from the right in a row of n = n " rank + 1; always convert to one consistent numbering before solving.
- Facing the center of a circle: clockwise = left, anticlockwise = right (reversed if facing outward).
- 'Left/right of' allows any distance; 'immediately left/right of' fixes strict adjacency — never conflate the two.
- In double rows facing each other, match physical (not viewer) position numbers across rows for 'opposite' pairs.
- Place absolute-position clues on the diagram first, then relative clues, then negative constraints last.
- Update a drawn diagram step by step rather than tracking placements mentally.

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IndiaBix — Seating Arrangement Questions

<https://www.indiabix.com/logical-reasoning/seating-arrangement/>

GeeksforGeeks — Logical Reasoning Questions and Answers

<https://www.geeksforgeeks.org/logical-reasoning-questions-and-answers/>